UMESH

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EDUCATION

B.TECH(MECHANICAL ENGINEERING)	2023-2027	NSUT, New Delhi	6.49
CBSE(Class XII)	2022	Sarvodaya Bal Vidyalaya, New Delhi	84%
CBSE(Class X)	2020	Sarvodaya Bal Vidyalaya, New Delhi	82%

PROJECTS

- Engineering Projects **Portfolio** – CAD models and simulations hosted online : LINK
- Designed mechanical systems including a Gantry Coordinate Measuring Machine **(CMM)** and **Geneva Mechanism** using **Fusion 360** with parametric modelling and assembly design.
- Modelled an approximate **truck body in SpaceClaim** and performed **aerodynamic CFD analysis** using **ANSYS Fluent / CFX**, including geometry preparation and meshing.
- Conducted **fluid flow** simulation of a **pipe elbow junction** with dual inlets (hot and cold water) and a single outlet to study flow behaviour and mixing characteristics.

ACADEMICS AND SKILLS

- Completed core **Mechanical Engineering subjects** including Manufacturing Processes, Theory of Machines, Engineering Mechanics, Thermal, Fluids and Control Systems.
- Completed a **CAE training program** covering **Fusion 360 and ANSYS**, gaining practical exposure to CAD modelling, FEM concepts, and basic CFD analysis.
- Continuously developing practical engineering skills through **hands-on CAD modelling, simulation projects, and technical learning** alongside coursework.

TECHNICAL SKILLS

- **CAD :** Fusion 360, SpaceClaim, NX (learning).
- **CAE :** ANSYS Workbench (Fluent, CFX, Structural, Thermal, Rigid dynamics)
- **Core Analysis :** Structural, Thermal, Fluid Flow, Topologies Optimization.

CERTIFICATION

CAE Certification : Industrial CAE Training program, NSUT — **09th June – 18th July 2025**

- Completed training covering **CAD modelling, FEA fundamentals, and CFD basics** using **Fusion 360 and ANSYS**.
- Attended technical sessions by industry professionals from **Autodesk, Curious3D, and ARK Infosolutions**.